An Explainable Automated Cyber-Insurance Underwriting and Continuous Monitoring Platform

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Abstract

Indian SMEs face the same ransomware, email-compromise and payment-fraud attacks as large firms, yet few carry cyber insurance, and the cover they can buy is priced from a self-completed questionnaire that nobody checks again. Beretta is an open prototype that links evidence, pricing, binding and post-bind monitoring in one inspectable pipeline. Evidence for each control is graded from DECLARED to TECHNICALLY_OBSERVED; a frequency–severity engine prices five perils from published claims data; and after bind a read-only agent sends signed reports that move warranty states but never the premium. Every parameter carries its source, and all 224 automated tests pass.

Problem Statement

1. **Priced with traditional thinking**
   Short, heavy-tailed loss history; premiums rest on revenue and industry.
2. **The premium is a black box**
   The insured cannot see which answer raised it or what would lower it.
3. **One questionnaire, once a year**
   A self-completed form fixes the price until renewal.
4. **Controls are never verified**
   A declared "yes" is priced the same as a test result.
5. **A static view of a dynamic landscape**
   Nothing is checked after bind; gaps surface only at claim time.

Market Context

8.84 crore
enterprises on India's Udyam portals (2026)

Under 5%
of Indian companies hold cyber cover (estimates)

18.4% a year
forecast growth in India, against 14.2% worldwide

₹25.5 crore
average cost of a breach in India (IBM 2026)

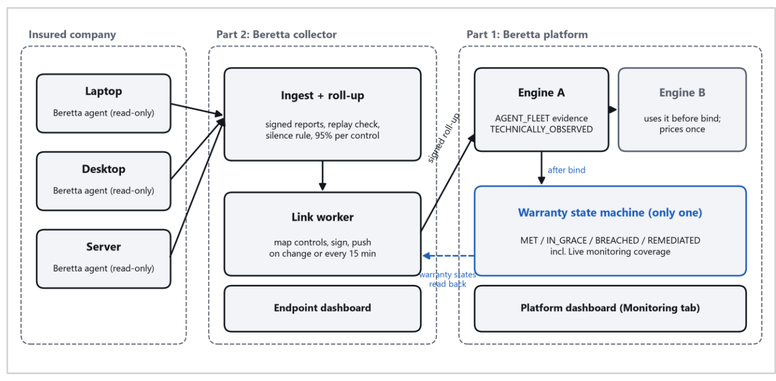
The gap: premium is concentrated in large firms, underwriting still rests on a self-completed questionnaire, and we did not find an Indian insurer that includes continuous monitoring in its policy. Sources: Ministry of MSME via KNN India; IJIRL (2025); MarketsandMarkets; IBM Cost of a Data Breach 2026.

Objectives

- Verify declared controls with a read-only agent
- Grade evidence by strength
- Develop a pricing mechanism from industry data
- Make premiums transparent and explainable
- Monitor warranties continuously after bind
- Give every user an interactive dashboard

Methodology

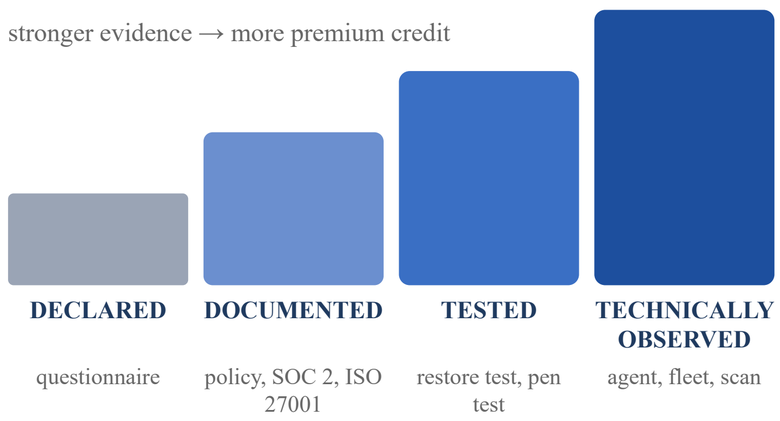
Two engines share one record of evidence. Signed reports flow from the insured's machines to the collector, and one signed roll-up per company flows to the platform.



System architecture

Risk engine: grading evidence by strength

stronger evidence → more premium credit

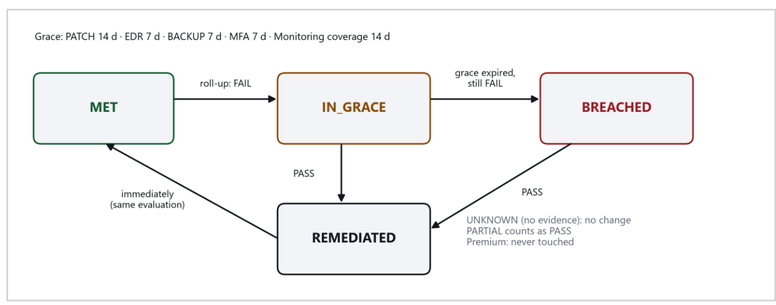


Pricing engine: five perils, priced once

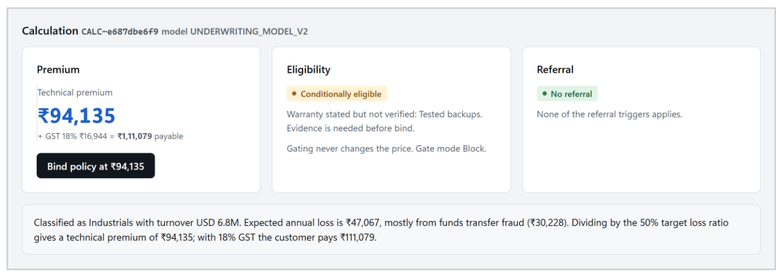
Frequency	$\lambda = \lambda_{global} \times turnover \times sector \times controls \times claims$
Severity	lognormal loss, $\sigma = 2.193$ (NetDiligence SME claims)
Payout	integrated through the retention and the limit
Premium	expected loss ÷ 0.50 target loss ratio, + 18% GST

A stated control earns 60% of a verified control's credit; claims history uses credibility ($\alpha = 4.22$). Every number carries its source and a tag.

After bind: warranties, never the premium



Every quote explains itself



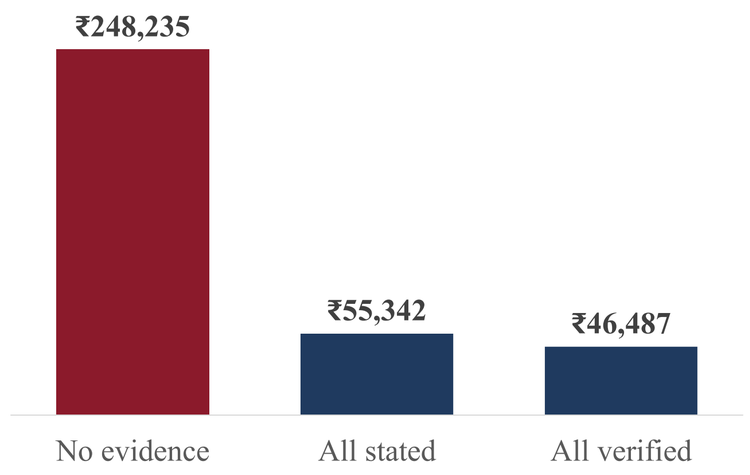
The quote shows the premium, what is needed before cover, and how the figure was reached.

Built with

Python standard library · HTML, CSS, JavaScript · SQLite · HMAC-SHA256 · TLS

Results

Technical premium for a reference firm (₹)



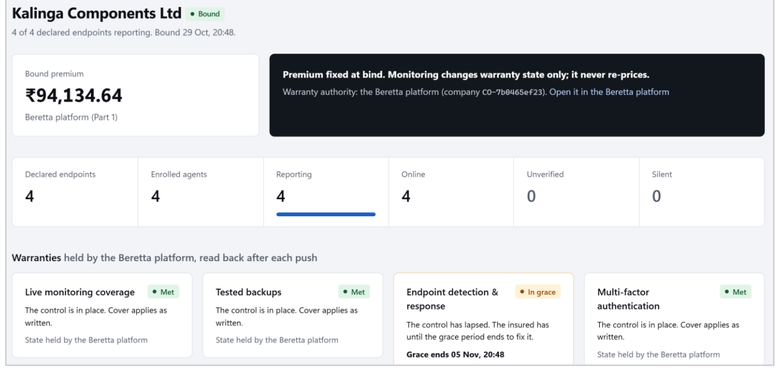
224 / 224
automated tests pass (65 platform, 158 agent and collector, 1 end to end)

₹32,421
saved by verifying privileged access alone (worked example)

1.38% vs 1.48%
modelled peril frequency vs published (1.47% with other claims)

Under 16%
largest premium change from any one assumption

Live demonstration (SIMULATED endpoints)



EDR stopped on 1 of 4 machines: the warranty moved to IN_GRACE within seconds, then BREACHED, then back to MET after the fix. The bound premium stayed at ₹94,135, and a re-price was refused.

Future Work

- A sturdier, more autonomous agent that tracks current threats such as zero-days, staying read-only
- More controls monitored, with identity-provider connectors to verify MFA
- Better loss models from data, including Indian claims
- Policy conditions and limits on the platform, so coverage and payouts appear on the dashboard
- macOS and wider OS support

References

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